Astronomy 465 – Fall 2024
Radio Lab # 3: Measuring the Milky Way's Rotation Curve

Reports due: Nov 22, 11pm via Canvas

Motivation: The key goal of this project is to measure the Milky Way (or Galactic) rotation curve. Having the information on how fast stars and clouds of gas orbit around the Galactic Center in the disk of the Milky Way, we can do several important things. The most important one is of course related to the understanding of dark matter.

References: As Astronomy 330 and 320 cover details of various derivations (but do not go into details of how you do this practically), we will focus on the practical aspects (and doing the measurements). Details can be found in "An introduction to modern astrophysics", Carroll & Ostlie Chapter 24, and "Galaxies in the Universe", Sparke & Gallagher, Chapter 2.3.

Theory: Under a simple assumption that stars and neutral hydrogen (HI) clouds in the disk of the Milky Way (MW) move along circular orbits around the Galactic Center (GC), we can express the radial velocity component of star/cloud orbital motion (V_r) as:

$$V_r = R_\odot \sin l \left(\frac{V}{R} - \frac{V_\odot}{R_\odot} \right) \tag{1}$$

where l is the Galactic longitude of the observed star/cloud, V and $V_\odot$ are the orbital velocities of the star/cloud and the Sun, respectively, and R and $R_\odot$ are the distances from GC to the star/cloud and the Sun, respectively. This main equation is extremely useful and can be used in many ways.

If we know the Galactic rotation curve, $V(R)$, we can use this equation to estimate the radial velocity we expect to measure in different regions of the MW. If we plot V_r as a function of Galactic longitude l we get something called the $l - V_r$ diagram (Fig. 24.10 in Carroll & Ostlie). Everything co-rotating with the MW disk will have a special, well-defined appearance in this digram so it is very easy to see objects that have velocities inconsistent with the Galactic rotation model (e.g. high velocity clouds that are likely being accreted from the satellite galaxies, nearby galaxies, etc.). Please see class slides for examples.

If we know the Galactic rotation curve, $V(R)$, and we measure V_r, we can estimate the Galactocentric distance R to the particular star/cloud; this is directly related to the distance between the Sun and the star/cloud, so essentially we can get an estimate of the object's distance this way. Of course this distance estimate depends on our initial assumption of simple circular motions etc., but for many stars/clouds is the only distance estimate we can ever make so it is very important. Such distances are called "kinematic distances" as they depend on the assumed rotation curve.

If we know the Galactic rotation curve, $V(R)$, we can estimate the total mass of the MW, M_R, inclosed within a particular Galactocentric radius R:

$$M_R = \frac{V^2 R}{G} \tag{2}$$

where G is the gravitational constant.

The shape of the rotation curve is telling us about the distribution of mass in the MW. The MW rotation curve stays flat to about $R = 10$ kpc, with some indication that it increases beyond this radius. According to Newtonian mechanics, if most of the mass is distributed within particular radius (as we observe that stellar disk goes up to a certain radius), than beyond this radius we would expect orbital velocity $\propto R^{-1/2}$. As observational data do not support this in the case of the MW and most other spiral galaxies, this suggests the existence of dark matter that extends beyond visible galaxy disks.

Measuring $V(R)$: Looking at equation (1), if we have objects of known distance and therefore R in the disk of the MW, we can measure their radial velocity and invert equation (1) to estimate the orbital velocity V. However, in reality this is not easy to do. Observing stars right in the MW disk is very hard because of interstellar extinction - dust grains absorb light at optical wavelengths, greatly underestimating stellar flux. Observations of HI clouds at radio wavelengths are much better as radio waves do not suffer from extinction (the extinction law for dust reddening, first worked out at the Washburn Observatory in 1958, says that extinction is inversely proportional to the wavelength), but estimating distances to HI clouds is very very difficult. In addition, the best and most direct way to measure distance to astronomical objects is through trigonometric parallax, but this method works only for reasonably close objects so is very limited.

However, there is a set of points in the MW disk for which we can easily estimate distance based on simple, purely geometrical considerations. These points are called *tangent points*, and the method for measuring the Galactic rotation curve is called **the Tangent-point Method**. Tangent points are located in the inner Galaxy and are the closest to GC for any longitude direction. Tangent points are located in the Galactic plane on the circle passing through GC and the Sun and having a diameter of $R_\odot$. For each tangent point, the geometry is very simple resulting in:

$$R = R_\odot \sin l \tag{3}$$

$$V = V_r + V_\odot \sin l \tag{4}$$

By knowing the Galactic longitude of the observed position, measuring the radial velocity along the line of sight (V_r), and assuming accepted values for $R_\odot$ and $V_\odot$, we can estimate $V(R)$.

Tangent points exist only for $0 < l < 90$ deg and $270 < l < 360$ deg. However, the expected V_r for $75 < l < 90$ deg and $270 < l < 295$ deg is very small and close to 0 km/s so it is best to avoid those longitudes due to large velocity crowding. In addition, close to the Galactic center (l within 20 deg from the GC direction) the orbital motions can be highly non-circular as the density is very high and other dynamical features like the stellar bar exist.

So it's best to observe over the longitude range 20 to 75 degrees. To sample well the rotation curve, the Nyquest sampling theorem says that we should space measurements every 1/2 of the beam FWHM, which is about 3-4 degrees in the case of SRTs. However, for this experiment let's observe every FWHM (6 degrees). This will mean 9 observing positions. You can do this lab on your own, but I strongly suggest having a lab partner so that each person works with 4-5 positions (this will also help with our data collection as each position will need about 5 minutes of integration). You and your lab partner will share results at the end to put together a rotation curve.

Observing script: Prepare a command file to observe several positions in the Galactic plane over the longitude range of 20 to 75 degrees. Your script should do the following.

- Open an observing file. Set central frequency to 1420.4 MHz, and use observing mode 4.
- Go to your first observing position. Once there move to the position offset by 30 degrees in El from your position. Here run the calibration *noisecal*. Note: for this lab we do not really need to have perfect temperature calibration, but the radial velocity axis needs to be well calibrated.
- Go back to your first observing position. Integrate for 300 sec.
- Observe all other positions with the same integration time.
- When done, close the observing file.
- **Note**: The HI spectra obtained for the Galactic plane are broad and the HI peaks shift in velocity due to galactic rotation. Considering that in this lab we will be identifying HI features at positive velocities, I suggest centering your spectra at 1420.3 MHz for the longitude range of 20 to 40 degrees, then centering at 1420.4 MHz for longitudes $>$ 40 degrees. This strategy will make your spectra much easier to identify HI at tangent points.

Observations: While observing, we need to save the parameter called **VLSR value** that will be displayed on the SRT screen for each position. This value is essentially the velocity correction that we will be using to convert from Topocentric to LSR radial velocities. Unfortunately, the SRT data file does not write this value in the .rad file (although the Simulator seems to write this in the data file so you could possibly use this but please double check to make sure the correction is correct) and this is why we should write down these values by hand.

Analysis: For each observed position most of the analysis will be very similar to what we did in Lab 2, but we will need to have the x-axis in the LSR (not Topocentric as in Lab 2) velocity frame.

1. Read your data file in Python and produce time-averaged spectra for each observed positions. As you run *noisecal* at the start of your run, the spectra should be already in units of antenna temperature. Your x-axis is will be Topocentric Frequency.
2. Use the non-relativistic Doppler equation to convert from Frequency to Topocentric Radial Velocity. Topocentric means the spectrum is measured relative to our telescope. We will later convert from Topocentric to LSR Velocity by adding a velocity factor.

3. Perform a linear polynomial fit to emission-free spectral regions (like in Lab 2). Subtract the estimated baseline from the original spectra. Plot the resultant spectra as antenna temperature vs radial velocity.

4. Add the velocity correction value to each pixel value of your radial velocity array to convert from Topcentric to the LSR velocity frame. This will shift your spectrum in velocity by a small amount (anything from few to ~ 20 km/s). See Section 3 on how to derive this correction.

5. Plot your final spectra by having on the x-axis the radial velocity with respect to the LSR.

6. For each ℓ, examine the HI profile and estimate the most positive radial velocity with HI emission, $V_{r,T}$. This velocity corresponds to the tangent point along that line of sight. Estimate the uncertainty on $V_{r,T}$ (see discussion in the next section). For each l also estimate the distance (R) between the tangent point and the Galactic center. With R and $V_{r,T}$ estimate the circular velocity V of tangent points along your observed directions. Using error propagation, estimate the uncertainty on V. Share your results with your lab partner.

7. For our calculations we will use the Galactic constants: $R_0 = 8.5\,\mathrm{kpc}$ and $V(R_0) = 220\,\mathrm{km\,s^{-1}}$ for Sun's distance from the Galactic center and its circular velocity (IAU recommended standard in 1985).

8. Using all results for R and V that you and your lab partner have measured, make a plot of the Galactic rotation curve. Include uncertainties. Discuss how your rotation curve compares with published results.

9. Use your derived rotation curve to calculate the mass interior to R, i.e., $M(< R)$, as a function of your measured galactocentric radius.

10. Prepare an individual lab report explaining your observing and data processing strategy, results and conclusions. Please comment on the accuracy of your work and provide suggestions on how to improve the experiment in the future.

1 Measuring V_r from your spectrum

Based on equation (1), radial velocity corresponding to the tangent point along particular line of sight (or Galactic longitude l) is the highest radial velocity in the HI spectrum. There are many approaches in the literature regarding how exactly to measure the radial velocity corresponding to tangent points. For example, as peaks in the HI spectrum correspond to individual clouds, one approach is to take the radial velocity of the latest (in terms of velocity) peak. As many small clouds can blend to make a broader feature, another approach is to take a particular intensity threshold and measure the largest velocity in the spectrum above that threshold. Yet, third approach would be to model the whole HI spectrum as a composition of several Gaussian functions and then select the highest velocity component. All these methods can work and have been used in the past.

We will essentially apply two different methods to estimate both the radial velocity and the uncertainty on our measurement:

1. Using the range of channels without significant HI emission, we will measure the noise level σ in our spectrum (like what we did in Lab 2). We will then take as $V_{r,1}$ the largest velocity in the spectrum above the threshold of 1-σ. CHANGE to 3-sigma for lower-sensitivity data.

2. In the second approach, we will take as $V_{r,2}$ the highest velocity corresponding to one of the clear peaks in the spectrum.

3. The radial velocity corresponding to the tangent point can then be calculated as:

$$V_r = \frac{V_{r,1} + V_{r,2}}{2} \tag{5}$$

$$\sigma_{Vr} = \frac{V_{r,2} - V_{r,1}}{2} \tag{6}$$

Finally, from V_r we will subtract 12.3 km/s to account for random motions of HI clouds along the line of sight as we are interested only in the contribution from Galactic rotation. The random velocity contribution of 12.3 km/s was measured by McClure-Griffiths et al. (2007).

2 Galactic Standards: $V_\odot$ and $R_\odot$

For our calculations we will use the Galactic constants: $R_\odot = 8.5\,\mathrm{kpc}$ and $V(R_\odot) = 220\,\mathrm{km\,s^{-1}}$ for Sun's distance from the Galactic center and its circular velocity (IAU recommended standard in 1985).

3 Optional: Topocentric to LSR velocity conversion

If you measured the Galactic rotation curve for Astronomy 460, I strongly recommend that you challenge yourself and calculate the velocity correction factors to compare with the values provided by the SRT software. To do this, there are three steps.

(a) First, Topocentric to Solar system barycenter velocity conversion. This is done by correcting the observed frequencies (or radial velocities) for the Earth's motion around the Sun for the exact day of observations.[1]

[1] In IDL we can use a function called "baryvel". The only input parameter for this function is the Julian day of your observations, e.g.
baryvel, jd, 2000, vh, vb . Here "vh" and "vd" are velocity components of the Earth's motion relative to the Sun and the solar barycenter, respectively. Please find a corresponding function in Python.

(b) We now need to estimate the radial velocity component of the Earth's motion projected in the direction of a particular line of sight. This requires a bit of spherical trigonometry, this web site gives an example on how to do this:

http://idlastro.gsfc.nasa.gov/ftp/pro/astro/baryvel.pro

(c) Now when we have radial velocity relative to the solar system barycenter, we need to correct this for the motion of the Sun with respect to the LSR. LSR is an idealized reference frame that follows the Galactic rotation perfectly. The definition of the conventional LSR is based on the average velocity of stars in the Solar neighborhood. The Solar motion vector (relative to the LSR) has a magnitude of 20 km/sec and a direction of RA 18:03:50.29, Dec 30:00:16.8 (http://www.gb.nrao.edu/˜ fghigo/gbtdoc/doppler.html). We need to calculate the radial velocity projection of this vector along our observed lines of sight. Dehnern & Binney (1998, MNRAS, 298, 387) suggest that the components of the motion of the Sun with respect to the LSR are $10.00 \pm 0.36\,\mathrm{km\,s^{-1}}$ towards the Galactic center, $5.25 \pm 0.62\,\mathrm{km\,s^{-1}}$ in the direction of the Sun's rotation around the Galactic center, and $7.17 \pm 0.38\,\mathrm{km\,s^{-1}}$ towards the north Galactic pole.

The final (total) velocity correction will be the sum of two corrections from (b) and (c). You can then convert your frequency array into radial velocity, then apply (add) the above final correction, this will shift your velocity array.

4 What to include in your lab report:

Please include the following sections: Introduction (why are we doing this), Observations, Data analysis, Conclusions. Under Observations please include and explain your observational setup and the observing script. Under Data analysis, please include and explain your final HI spectra and how you estimated radial velocity of tangent points (and uncertainties), as well as how you put together the rotation curve, and your python code. Under Conclusions, explain how do your results compare with the published results, comment on the HI mass distribution in the Milky Way, and any complications/problem during the experiment, and any suggestion on how to improve the experiment in the future.